Generalizing CDCL with Graph Backtracking

SAT Presentation

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Acknowledgements

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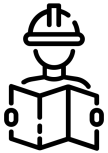




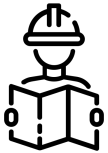


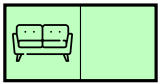


Introduction



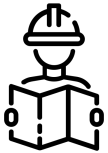
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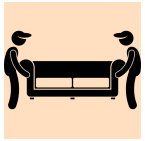


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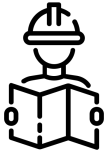


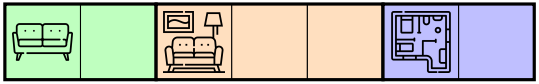




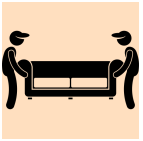


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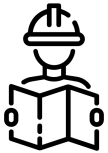


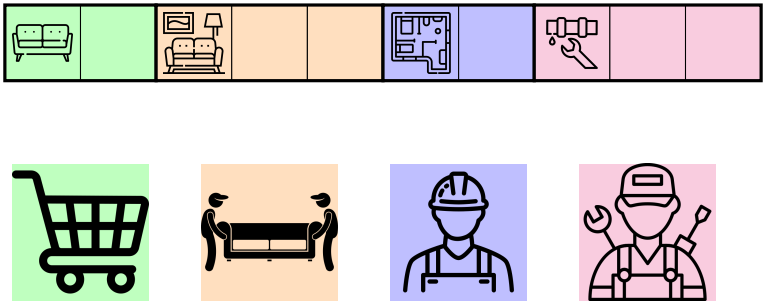




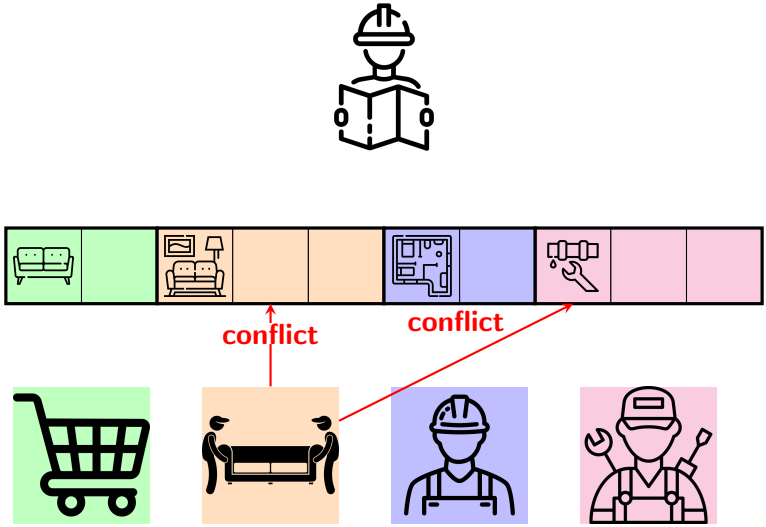


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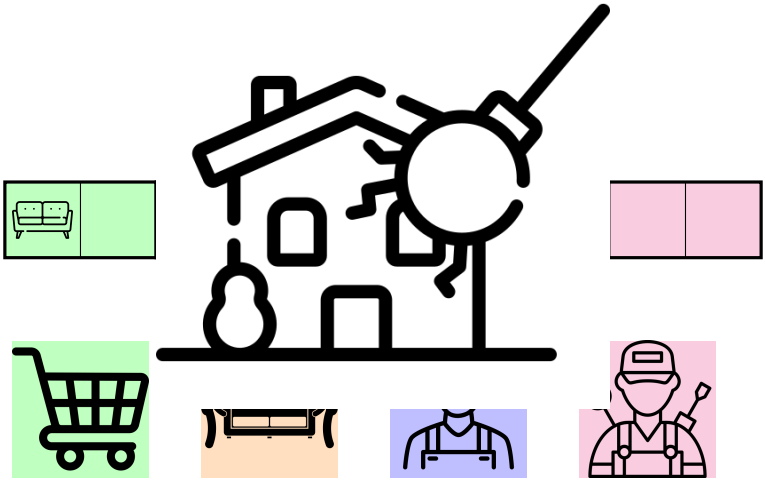




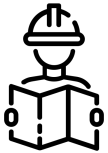
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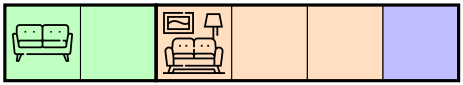


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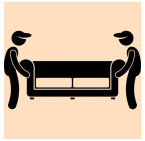


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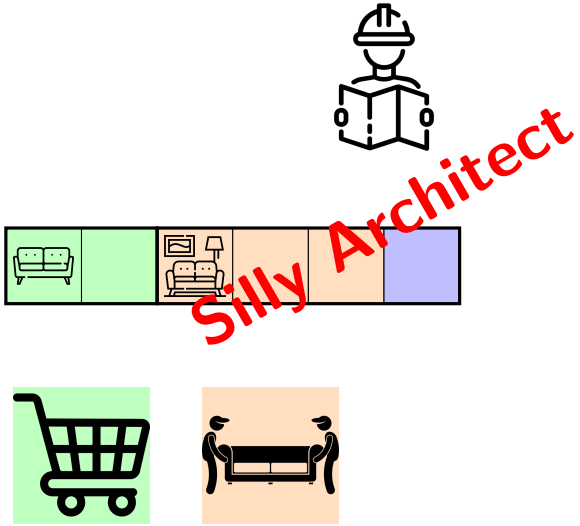






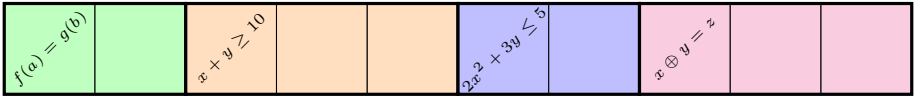


Introduction



SMT Architecture

SAT Solver



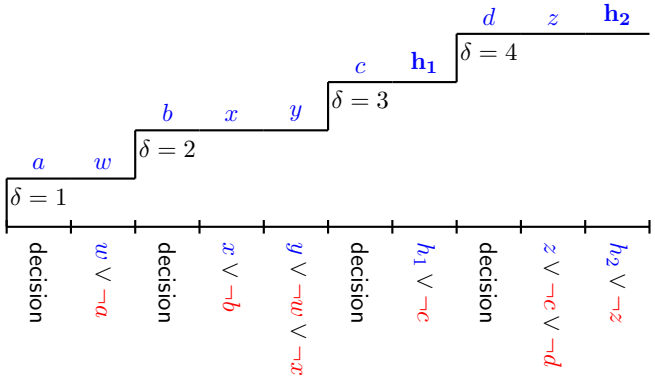
UF
(Uninterpreted Functions)

LRA
(Linear Real Arithmetic)

NRA
(Non-linear Real Arithmetic)

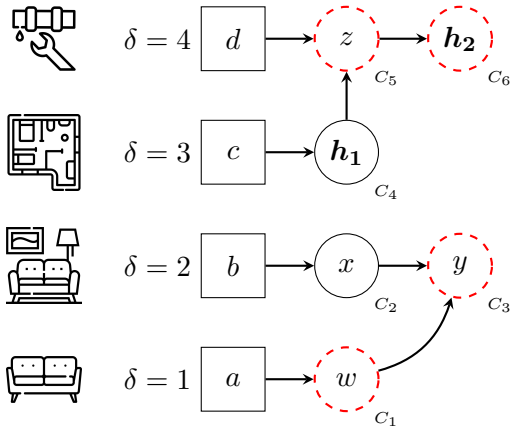
BV
(Bit Vectors)

Non-Chronological Backtracking – Trail



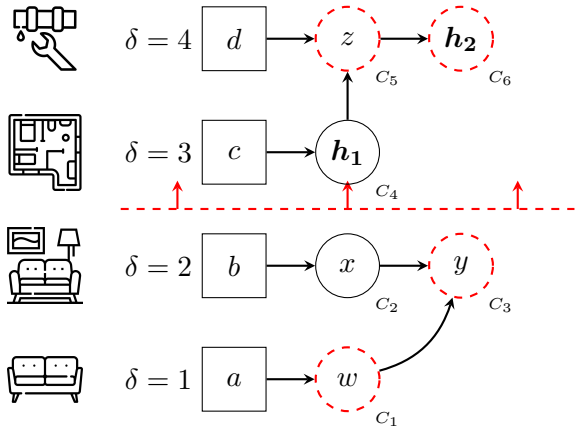
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Non-Chronological Backtracking – Implication Graph



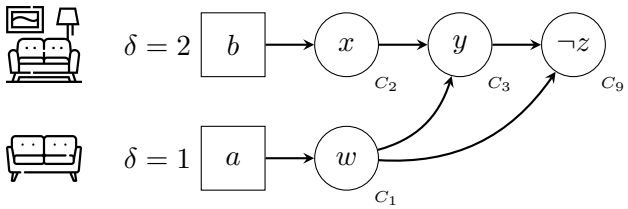
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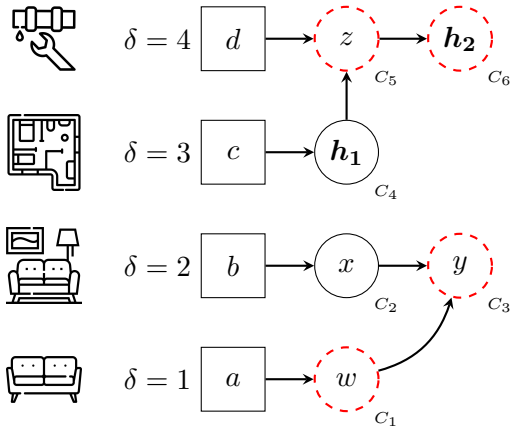
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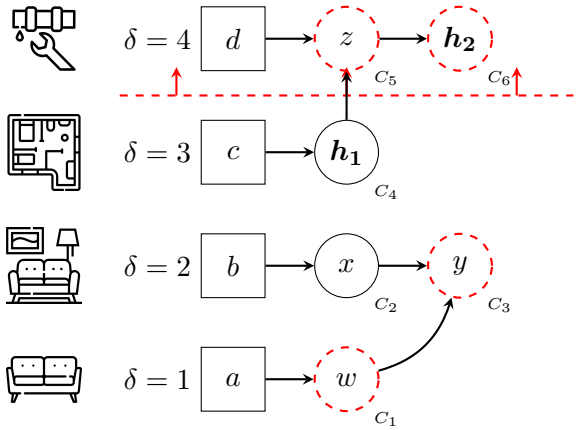
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Chronological Backtracking [NR18, MB19, Nad22, CFK24]



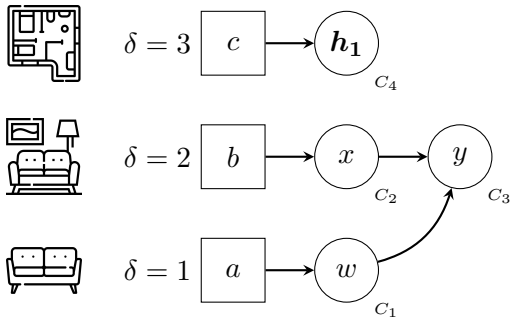
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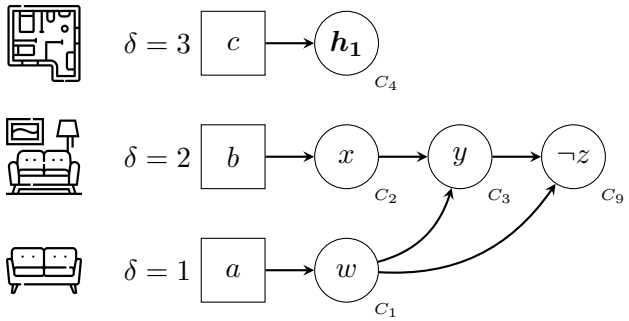
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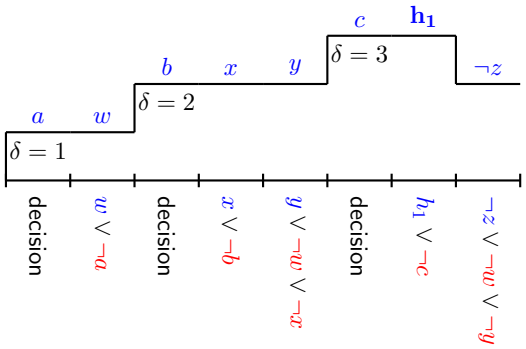
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To be fast

- NCB maintains nice **invariants**
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To be general

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To be smart

- Aggressive backjumping can undo a lot of useless work
- Maybe learn fewer clauses

A More Surgical Architect

Extended Interface

We allow the user to provide a **cost** for each literal.

A More Surgical Architect

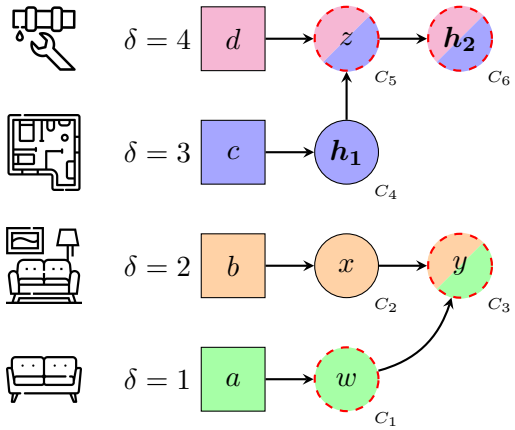
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New Backtracking Scheme

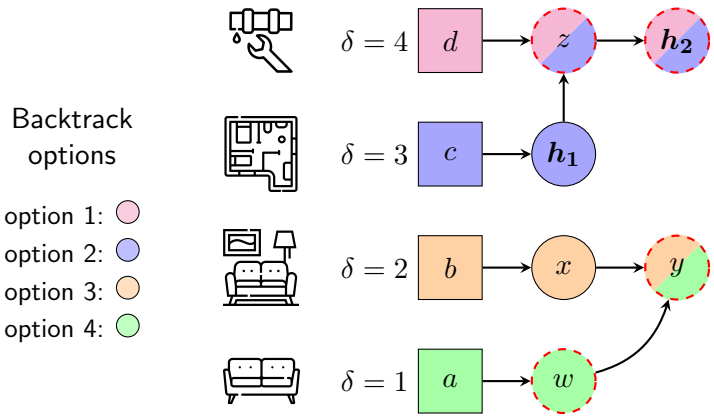
Graph Backtracking (GB) searches the cheapest set of literals to backtrack to resolve a conflict. We use the implication graph to find such a set.

Graph Backtracking (GB)

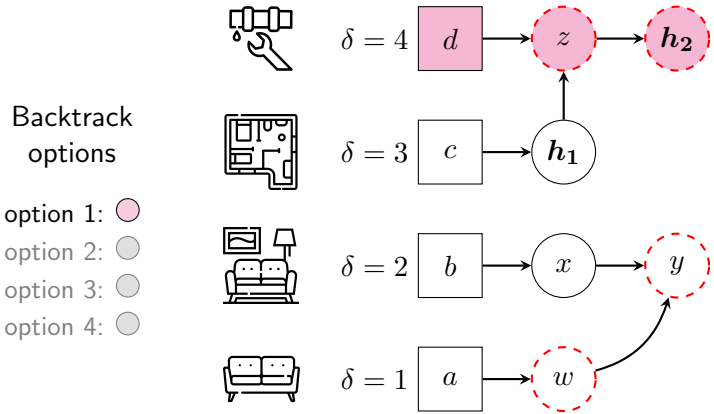


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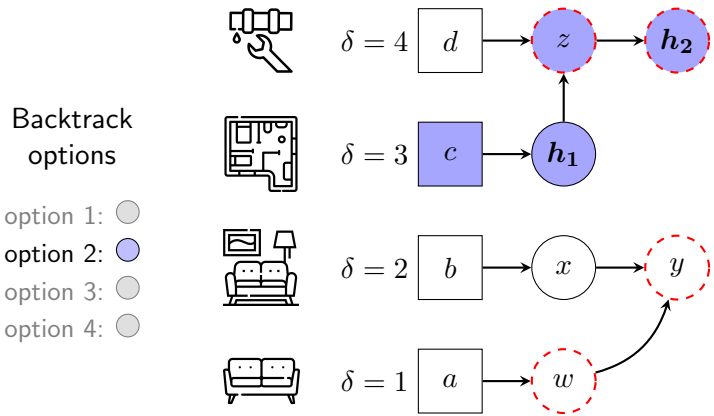
Graph Backtracking (GB)



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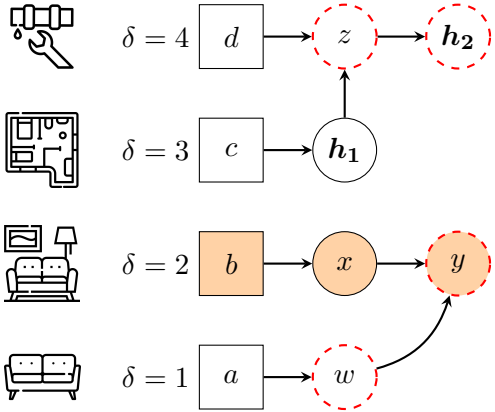
Backtrack options

option 1:

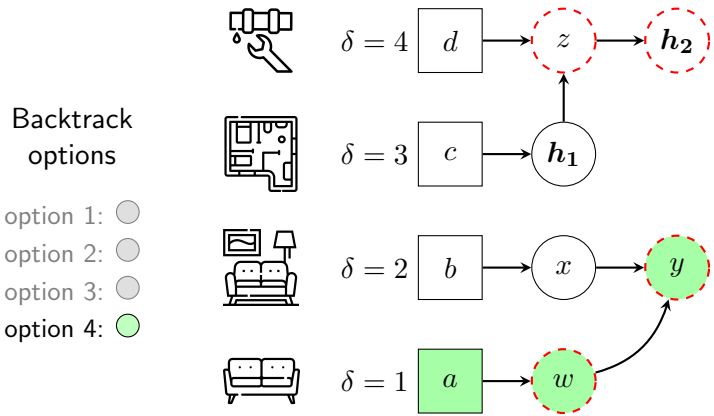
option 2:

option 3:

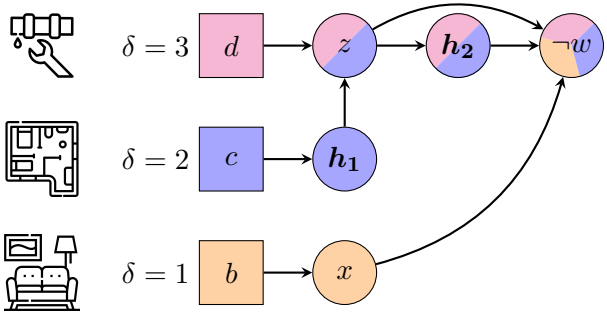
option 4:



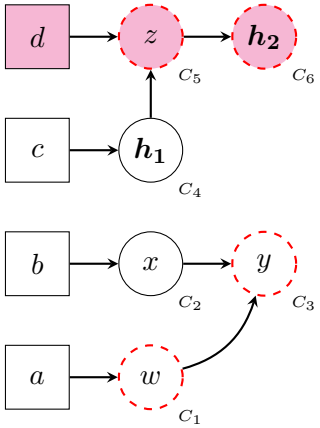
Graph Backtracking (GB)



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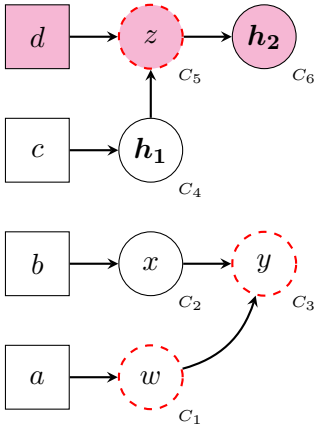
First Unique Implication Point (UIP) with GB



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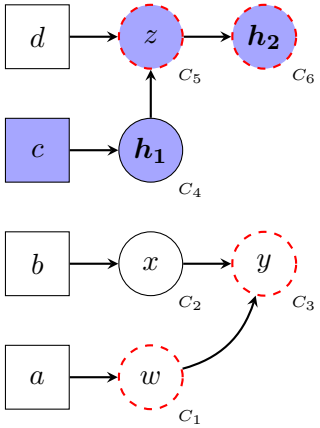
$\neg w \vee \neg y \vee \neg z$



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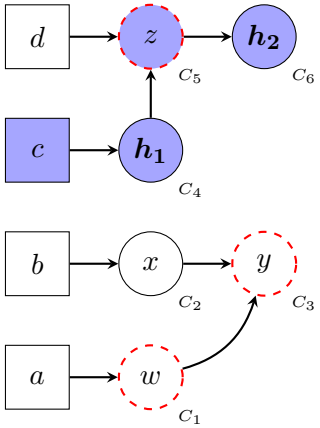


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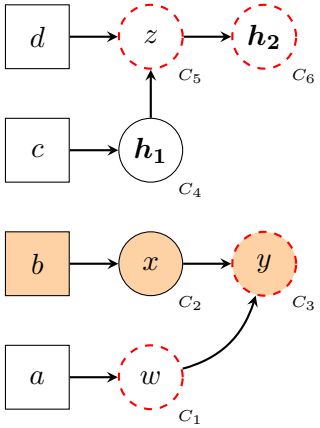
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First Unique Implication Point (UIP) with GB

$\neg w \vee \neg y \vee \neg z$

$\neg w \vee \neg y \vee \neg z$

$\neg w \vee \neg y \vee \neg z \vee \neg h_2$



$C_1 = w \vee \neg a$

$C_2 = x \vee \neg b$

$C_3 = y \vee \neg w \vee \neg x$

$C_4 = h_1 \vee \neg c$

$C_5 = z \vee \neg d \vee \neg h_1$

$C_6 = h_2 \vee \neg z$

$C_7 = a \vee \neg c \vee \neg d$

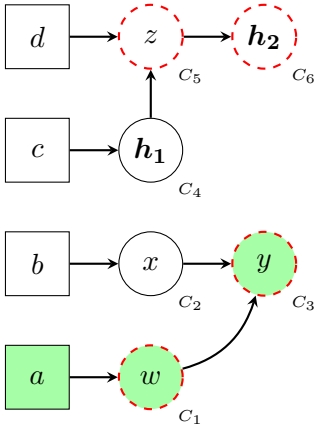
$C_8 = \neg w \vee \neg y \vee \neg z \vee \neg h_2$

First Unique Implication Point (UIP) with GB

$\neg w \vee \neg y \vee \neg z$

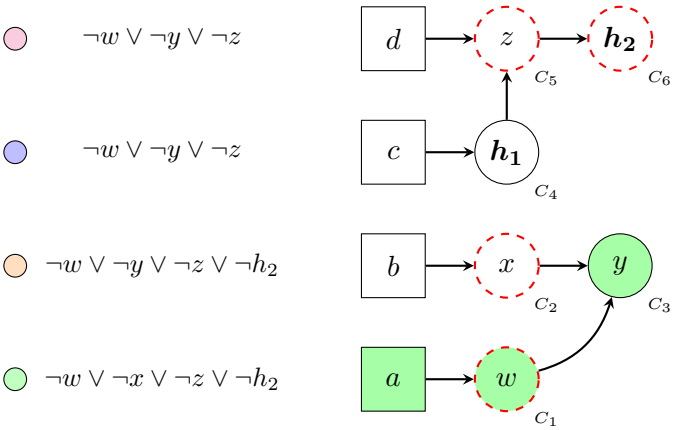
$\neg w \vee \neg y \vee \neg z$

$\neg w \vee \neg y \vee \neg z \vee \neg h_2$



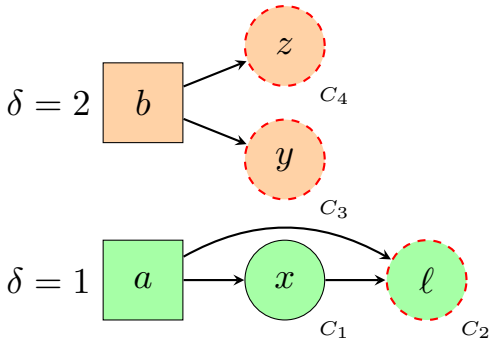
$$C_1 = w \vee \neg a$$
$$C_2 = x \vee \neg b$$
$$C_3 = y \vee \neg w \vee \neg x$$
$$C_4 = h_1 \vee \neg c$$
$$C_5 = z \vee \neg d \vee \neg h_1$$
$$C_6 = h_2 \vee \neg z$$
$$C_7 = a \vee \neg c \vee \neg d$$
$$C_8 = \neg w \vee \neg y \vee \neg z \vee \neg h_2$$

First Unique Implication Point (UIP) with GB



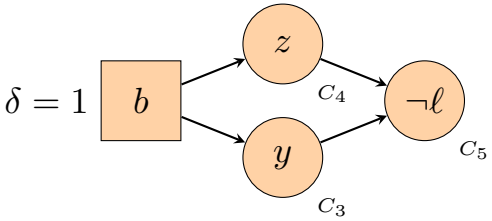
$$C_1 = w \vee \neg a$$
$$C_2 = x \vee \neg b$$
$$C_3 = y \vee \neg w \vee \neg x$$
$$C_4 = h_1 \vee \neg c$$
$$C_5 = z \vee \neg d \vee \neg h_1$$
$$C_6 = h_2 \vee \neg z$$
$$C_7 = a \vee \neg c \vee \neg d$$
$$C_8 = \neg w \vee \neg y \vee \neg z \vee \neg h_2$$

Challenge I: Termination



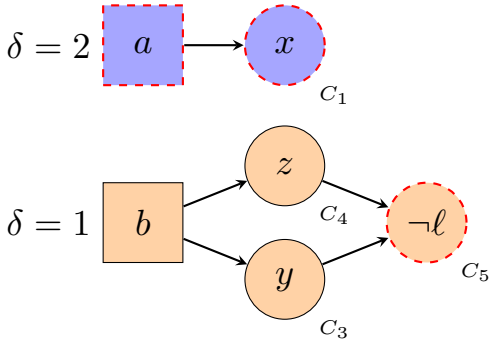
$$
\begin{aligned}
C_1 &= x \vee \neg a \\
C_2 &= \ell \vee \neg a \vee \neg x \\
C_3 &= y \vee \neg b \\
C_4 &= z \vee \neg b \\
C_5 &= \neg \ell \vee \neg y \vee \neg z
\end{aligned}
$$

Challenge I: Termination



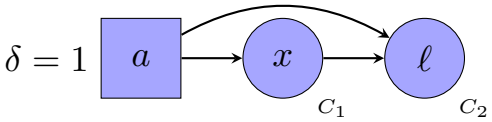
$$
\begin{aligned}
C_1 &= x \vee \neg a \\
C_2 &= \ell \vee \neg a \vee \neg x \\
C_3 &= y \vee \neg b \\
C_4 &= z \vee \neg b \\
C_5 &= \neg\ell \vee \neg y \vee \neg z
\end{aligned}
$$

Challenge I: Termination



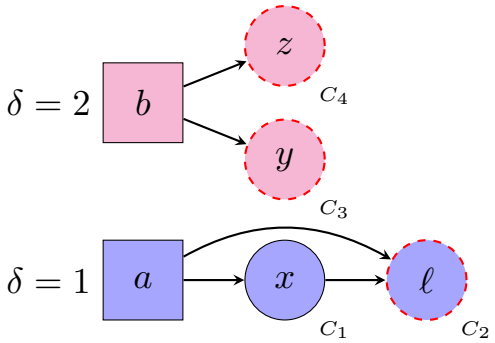
$$
\begin{aligned}
C_1 &= x \vee \neg a \\
C_2 &= \ell \vee \neg a \vee \neg x \\
C_3 &= y \vee \neg b \\
C_4 &= y \vee \neg b \\
C_5 &= \neg\ell \vee \neg y \vee \neg z
\end{aligned}
$$

Challenge I: Termination



$$
\begin{aligned}
C_1 &= x \vee \neg a \\
C_2 &= \ell \vee \neg a \vee \neg x \\
C_3 &= y \vee \neg b \\
C_4 &= z \vee \neg b \\
C_5 &= \neg \ell \vee \neg y \vee \neg z
\end{aligned}
$$

Challenge I: Termination



$$
\begin{aligned}
C_1 &= x \vee \neg a \\
C_2 &= \ell \vee \neg a \vee \neg x \\
C_3 &= y \vee \neg b \\
C_4 &= z \vee \neg b \\
C_5 &= \neg \ell \vee \neg y \vee \neg z
\end{aligned}
$$

Challenge I: Safeguards

Why this happens

- Arbitrary cost function
- No guarantee to learn a new clause after backtracking (like in CB [MB19])

Challenge I: Safeguards

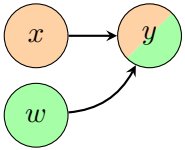
Why this happens

- Arbitrary cost function
- No guarantee to learn a new clause after backtracking (like in CB [MB19])

Solution

- If possible to learn a clause: choose the color that allows to learn the clause with minimum total cost
- Otherwise, select latest decision (like in CB)

Challenge II: Watched Literals



Let us watch y and $\neg x$ in C_3.

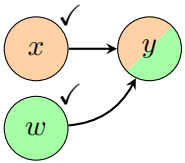
$$C_3 = \underline{y} \vee \underline{\neg x} \vee \neg w$$

Invariant (Watched Literals)

For each clause $C \in F$ watched by c_1, c_2, we have

$$\textit{propagated}(\neg c_1) \Rightarrow \textit{satisfied}(c_2)$$

Challenge II: Watched Literals



$C_3 = \underline{y} \vee \underline{\neg x} \vee \neg w$

Let us watch y and $\neg x$ in C_3.

$$\text{propagated}(\neg y) \Rightarrow \text{satisfied}(\neg x)$$
$$\text{propagated}(x) \Rightarrow \text{satisfied}(y)$$

is satisfied.

Invariant (Watched Literals)

For each clause $C \in F$ watched by c_1, c_2, we have

$$\textit{propagated}(\neg c_1) \Rightarrow \textit{satisfied}(c_2)$$

Challenge II: Watched Literals



Let us watch y and $\neg x$ in C_3.

$$\text{propagated}(\neg y) \Rightarrow \text{satisfied}(\neg x)$$
$$\text{propagated}(x) \Rightarrow \text{satisfied}(y)$$

is violated

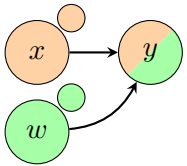
$$C_3 = \underline{y} \vee \underline{\neg x} \vee \neg w$$

Invariant (Watched Literals)

For each clause $C \in F$ watched by c_1, c_2, we have

$$\textit{propagated}(\neg c_1) \Rightarrow \textit{satisfied}(c_2)$$

Challenge II: Watched Literals (Solution)



We define cross-colors(ℓ) as the set of colors that trigger the repropagation of ℓ.

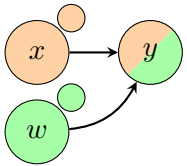
$$C_3 = \underline{y} \vee \underline{\neg x} \vee \neg w$$

Invariant (Watched Literals)

For each clause $C \in F$ watched by c_1, c_2, we have

$$\textit{propagated}(\neg c_1) \Rightarrow \big[\textit{satisfied}(c_2) \wedge \textit{colors}(c_2) \subseteq \textit{cross-colors}(c_1)\big]$$

Challenge II: Watched Literals (Solution)



$C_3 = \underline{y} \vee \underline{\neg x} \vee \neg w$

Let us watch y and $\neg x$ in C_3.

propagated$(\neg y) \Rightarrow$ [satisfied$(\neg x) \wedge$ ◯ $\subseteq$ ◔]

propagated$(x) \Rightarrow$ [satisfied$(y) \wedge$ ◔ $\subseteq$ ◯]

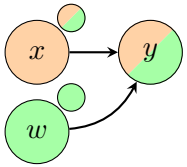
is satisfied.

Invariant (Watched Literals)

For each clause $C \in F$ watched by c_1, c_2, we have

$$\textit{propagated}(\neg c_1) \Rightarrow \left[\textit{satisfied}(c_2) \wedge \textit{colors}(c_2) \subseteq \textit{cross-colors}(c_1)\right]$$

Challenge II: Watched Literals (Solution)



$C_3 = \underline{y} \vee \underline{\neg x} \vee \neg w$

Let us watch y and $\neg x$ in C_3.

propagated$(\neg y) \Rightarrow$ [satisfied$(\neg x) \wedge$ ◯ $\subseteq$ ◯]

propagated$(x) \Rightarrow$ [satisfied$(y) \wedge$ ◯ $\subseteq$ ◯]

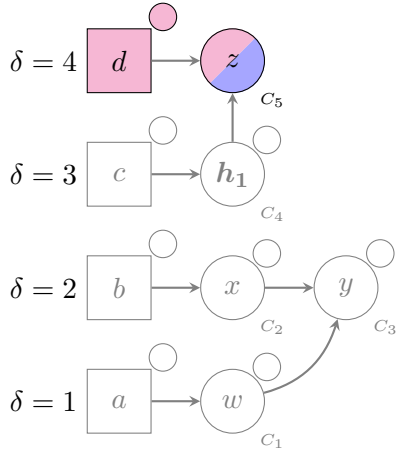
is satisfied.

Invariant (Watched Literals)

For each clause $C \in F$ watched by c_1, c_2, we have

$$\textit{propagated}(\neg c_1) \Rightarrow \left[\textit{satisfied}(c_2) \wedge \textit{colors}(c_2) \subseteq \textit{cross-colors}(c_1)\right]$$

Challenge II: Cross-Coloring Example



$C_1 = \underline{w} \vee \underline{\neg a}$

$C_2 = \underline{x} \vee \underline{\neg b}$

$C_3 = \underline{y} \vee \neg w \vee \underline{\neg x}$

$C_4 = \underline{h_1} \vee \underline{\neg c}$

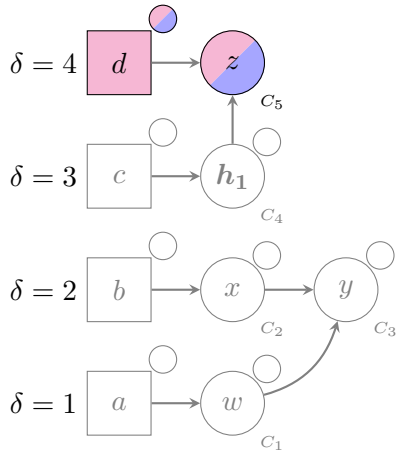
$C_5 = \underline{z} \vee \underline{\neg d} \vee \neg h_1$

$C_6 = \underline{h_2} \vee \underline{\neg z}$

$C_7 = \underline{a} \vee \neg c \vee \underline{\neg d}$

$C_8 = \neg w \vee \neg y \vee \underline{\neg z} \vee \underline{\neg h_2}$

Challenge II: Cross-Coloring Example



$C_1 = \underline{w} \vee \underline{\neg a}$

$C_2 = \underline{x} \vee \underline{\neg b}$

$C_3 = \underline{y} \vee \neg w \vee \underline{\neg x}$

$C_4 = \underline{h_1} \vee \underline{\neg c}$

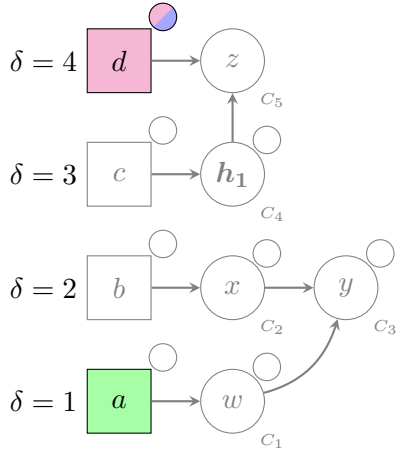
$C_5 = \underline{z} \vee \underline{\neg d} \vee \neg h_1$

$C_6 = \underline{h_2} \vee \underline{\neg z}$

$C_7 = \underline{a} \vee \neg c \vee \underline{\neg d}$

$C_8 = \neg w \vee \neg y \vee \underline{\neg z} \vee \underline{\neg h_2}$

Challenge II: Cross-Coloring Example



$C_1 = \underline{w} \vee \underline{\neg a}$

$C_2 = \underline{x} \vee \underline{\neg b}$

$C_3 = \underline{y} \vee \neg w \vee \underline{\neg x}$

$C_4 = \underline{h_1} \vee \underline{\neg c}$

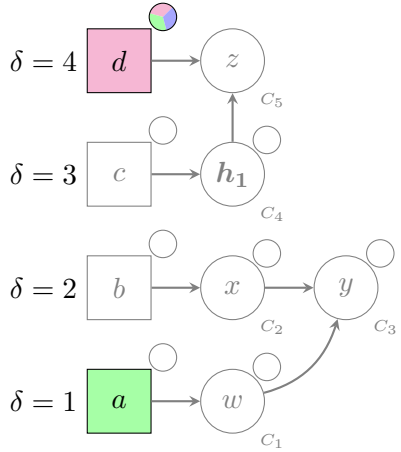
$C_5 = \underline{z} \vee \underline{\neg d} \vee \neg h_1$

$C_6 = \underline{h_2} \vee \underline{\neg z}$

$C_7 = \underline{a} \vee \neg c \vee \underline{\neg d}$

$C_8 = \neg w \vee \neg y \vee \underline{\neg z} \vee \underline{\neg h_2}$

Challenge II: Cross-Coloring Example



$C_1 = \underline{w} \vee \underline{\neg a}$

$C_2 = \underline{x} \vee \underline{\neg b}$

$C_3 = \underline{y} \vee \neg w \vee \underline{\neg x}$

$C_4 = \underline{h_1} \vee \underline{\neg c}$

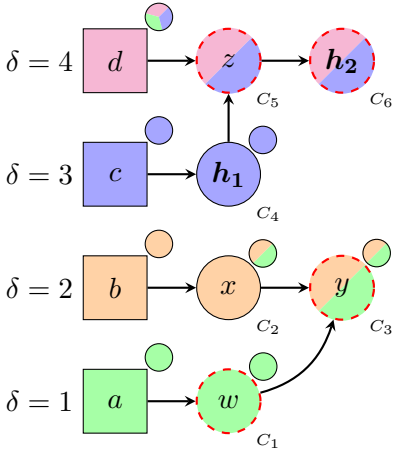
$C_5 = \underline{z} \vee \underline{\neg d} \vee \neg h_1$

$C_6 = \underline{h_2} \vee \underline{\neg z}$

$C_7 = \underline{a} \vee \neg c \vee \underline{\neg d}$

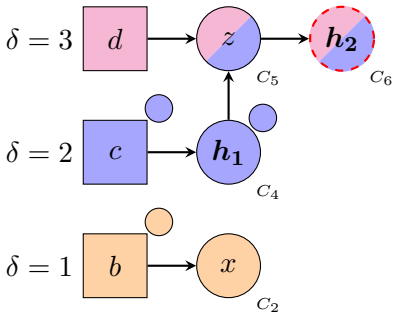
$C_8 = \neg w \vee \neg y \vee \underline{\neg z} \vee \underline{\neg h_2}$

Challenge II: Cross-Coloring Example



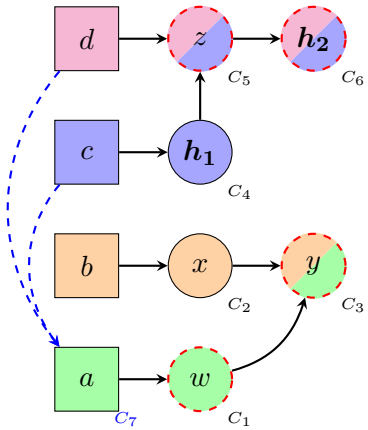
$$C_1 = \underline{w} \vee \underline{\neg a}$$
$$C_2 = \underline{x} \vee \underline{\neg b}$$
$$C_3 = \underline{y} \vee \neg w \vee \underline{\neg x}$$
$$C_4 = \underline{h_1} \vee \underline{\neg c}$$
$$C_5 = \underline{z} \vee \underline{\neg d} \vee \neg h_1$$
$$C_6 = \underline{h_2} \vee \underline{\neg z}$$
$$C_7 = \underline{a} \vee \neg c \vee \underline{\neg d}$$
$$C_8 = \neg w \vee \neg y \vee \underline{\neg z} \vee \underline{\neg h_2}$$

Challenge II: Cross-Coloring Example



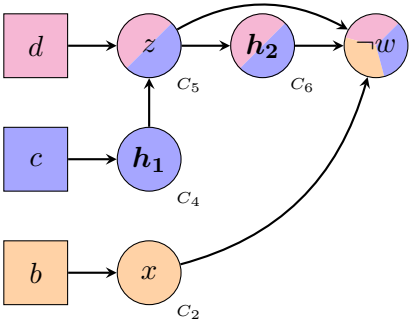
$$C_1 = \underline{w} \vee \underline{\neg a}$$
$$C_2 = \underline{x} \vee \underline{\neg b}$$
$$C_3 = \underline{y} \vee \neg w \vee \underline{\neg x}$$
$$C_4 = \underline{h_1} \vee \underline{\neg c}$$
$$C_5 = \underline{z} \vee \underline{\neg d} \vee \neg h_1$$
$$C_6 = \underline{h_2} \vee \underline{\neg z}$$
$$C_7 = \underline{a} \vee \neg c \vee \underline{\neg d}$$
$$C_8 = \neg w \vee \neg y \vee \underline{\neg z} \vee \underline{\neg h_2}$$
$$C_9' = \neg w \vee \neg x \vee \neg z \vee \neg h_2$$

Refinement: Color Merging



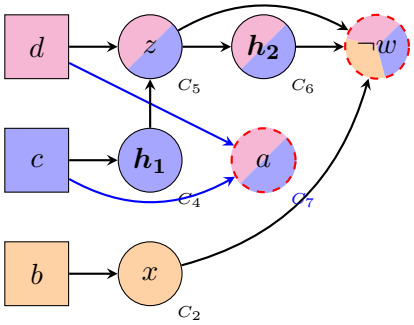
$$C_1 = w \vee \neg a$$
$$C_2 = x \vee \neg b$$
$$C_3 = y \vee \neg w \vee \neg x$$
$$C_4 = h_1 \vee \neg c$$
$$C_5 = z \vee \neg d \vee \neg h_1$$
$$C_6 = h_2 \vee \neg z$$
$$C_7 = a \vee \neg c \vee \neg d$$
$$C_8 = \neg w \vee \neg y \vee \neg z \vee \neg h_2$$

Refinement: Color Merging



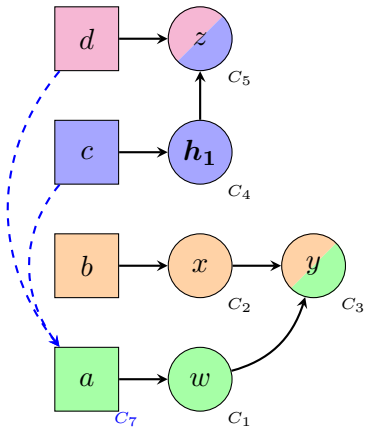
$$C_1 = w \vee \neg a$$
$$C_2 = x \vee \neg b$$
$$C_3 = y \vee \neg w \vee \neg x$$
$$C_4 = h_1 \vee \neg c$$
$$C_5 = z \vee \neg d \vee \neg h_1$$
$$C_6 = h_2 \vee \neg z$$
$$C_7 = a \vee \neg c \vee \neg d$$
$$C_8 = \neg w \vee \neg y \vee \neg z \vee \neg h_2$$
$$C_9' = \neg w \vee \neg x \vee \neg z \vee \neg h_2$$

Refinement: Color Merging



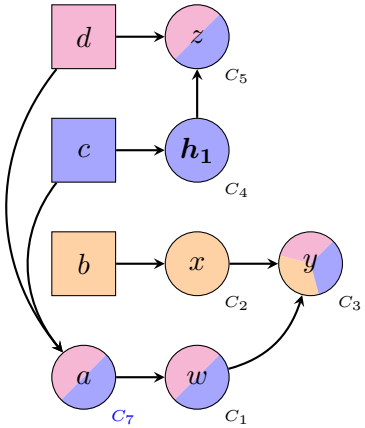
$$C_1 = w \vee \neg a$$
$$C_2 = x \vee \neg b$$
$$C_3 = y \vee \neg w \vee \neg x$$
$$C_4 = h_1 \vee \neg c$$
$$C_5 = z \vee \neg d \vee \neg h_1$$
$$C_6 = h_2 \vee \neg z$$
$$C_7 = a \vee \neg c \vee \neg d$$
$$C_8 = \neg w \vee \neg y \vee \neg z \vee \neg h_2$$
$$C_9' = \neg w \vee \neg x \vee \neg z \vee \neg h_2$$

Refinement: Eager Color Merging (ECM)



$$C_1 = w \vee \neg a$$
$$C_2 = x \vee \neg b$$
$$C_3 = y \vee \neg w \vee \neg x$$
$$C_4 = h_1 \vee \neg c$$
$$C_5 = z \vee \neg d \vee \neg h_1$$
$$C_6 = h_2 \vee \neg z$$
$$C_7 = a \vee \neg c \vee \neg d$$
$$C_8 = \neg w \vee \neg y \vee \neg z \vee \neg h_2$$

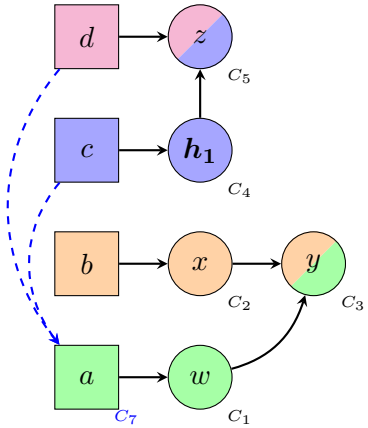
Refinement: Eager Color Merging (ECM)



$$C_1 = w \vee \neg a$$
$$C_2 = x \vee \neg b$$
$$C_3 = y \vee \neg w \vee \neg x$$
$$C_4 = h_1 \vee \neg c$$
$$C_5 = z \vee \neg d \vee \neg h_1$$
$$C_6 = h_2 \vee \neg z$$
$$C_7 = a \vee \neg c \vee \neg d$$
$$C_8 = \neg w \vee \neg y \vee \neg z \vee \neg h_2$$

Refinement: Lazy Color Merging (LCM)

Backtrack options



$$C_1 = w \vee \neg a$$
$$C_2 = x \vee \neg b$$
$$C_3 = y \vee \neg w \vee \neg x$$
$$C_4 = h_1 \vee \neg c$$
$$C_5 = z \vee \neg d \vee \neg h_1$$
$$C_6 = h_2 \vee \neg z$$
$$C_7 = a \vee \neg c \vee \neg d$$
$$C_8 = \neg w \vee \neg y \vee \neg z \vee \neg h_2$$

Refinement: Lazy Color Merging (LCM)

Backtrack options

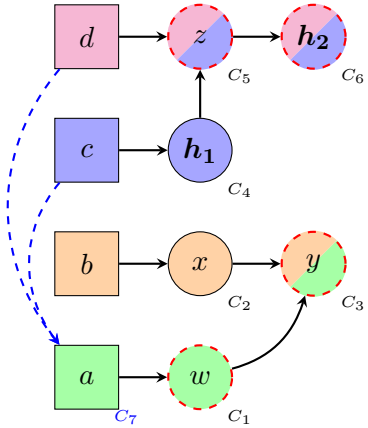
option 1:

option 2:

option 3:

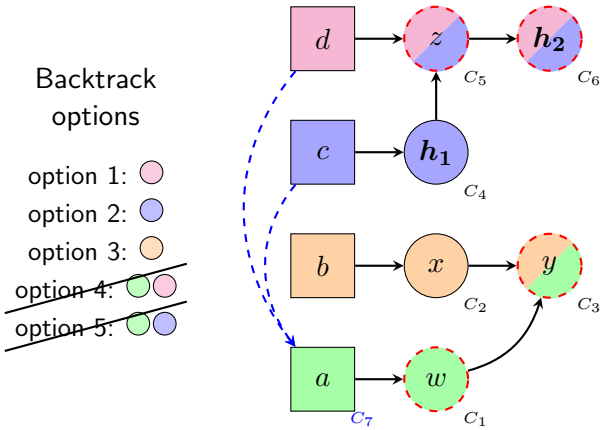
option 4:

option 5:



$$C_1 = w \vee \neg a$$
$$C_2 = x \vee \neg b$$
$$C_3 = y \vee \neg w \vee \neg x$$
$$C_4 = h_1 \vee \neg c$$
$$C_5 = z \vee \neg d \vee \neg h_1$$
$$C_6 = h_2 \vee \neg z$$
$$C_7 = a \vee \neg c \vee \neg d$$
$$C_8 = \neg w \vee \neg y \vee \neg z \vee \neg h_2$$

Refinement: Lazy Color Merging (LCM)



$$C_1 = w \vee \neg a$$
$$C_2 = x \vee \neg b$$
$$C_3 = y \vee \neg w \vee \neg x$$
$$C_4 = h_1 \vee \neg c$$
$$C_5 = z \vee \neg d \vee \neg h_1$$
$$C_6 = h_2 \vee \neg z$$
$$C_7 = a \vee \neg c \vee \neg d$$
$$C_8 = \neg w \vee \neg y \vee \neg z \vee \neg h_2$$

Preliminary Empirical Results

Option	# Timeout	PAR2	Time (s)		Propagations $\times 10^6$	
NCB	16	380.7	106 ± 428	-	23.7 ± 46.4	-
CB	11	333.1	103 ± 414	97.33%	22.9 ± 42.8	96.30%
GB	17	378.1	103 ± 402	96.91%	14.2 ± 25.3	60.05%
GB+ECM	11	299.7	72.4 ± 310	67.89%	12.6 ± 23.8	52.96%
GB+LCM	14	376.8	124 ± 485	117.1%	15.8 ± 29.9	66.39%

Table: Empirical results on 1000 satisfiable random 3-coloring problems with 650 nodes and 1469 edges with 2-hour timeout.

Reminder

- (N)CB: (Non-)Chronological Backtracking
- GB: Graph Backtracking
- ECM: Eager Color Merging
- LCM: Lazy Color Merging

Results in Vampire [BBC+25] with AVATAR [Vor14]

We ran Vampire's AVATAR with NapSAT on 10617 TPTP problems requiring AVATAR.

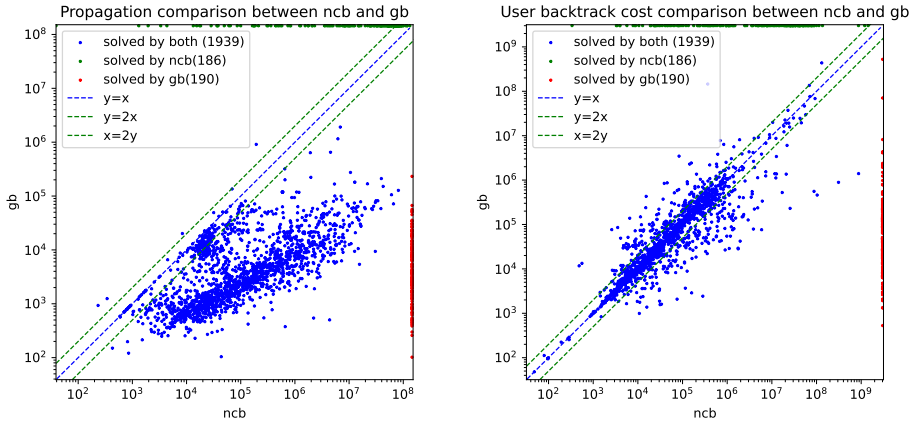


Figure: GB saves 98.7% of Boolean propagations and 70.3% of the estimated cost compared to NCB

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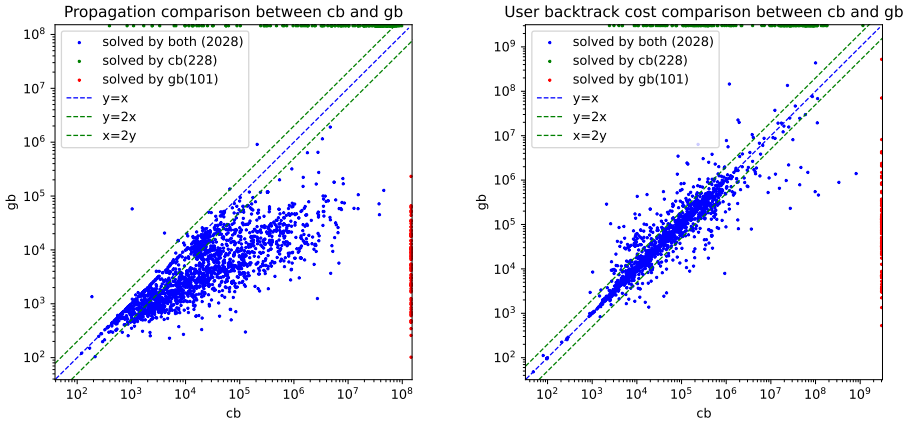
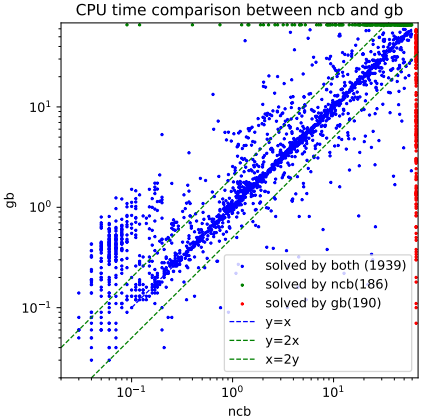
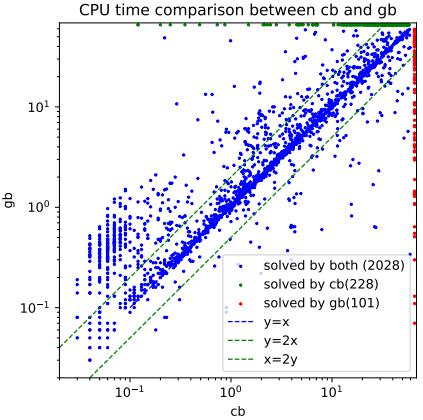


Figure: GB saves 95% of Boolean propagations and 69.3% of the estimated cost compared to CB

Results in Vampire [BBC+25] with AVATAR [Vor14]

We ran Vampire's AVATAR with NapSAT on 10617 TPTP problems requiring AVATAR.





Conclusion

Summary

- Graph Backtracking (GB), a new backtracking scheme for SAT solvers
- GB extends the SAT API with cost functions
- GB is implemented in NapSAT and integrated in Vampire's AVATAR
- GB performs significantly fewer Boolean propagations than (N)CB.
- GB achieves the goal of reducing the estimated cost of backtracking in AVATAR.

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Future Work

- Tune Vampire AVATAR's cost heuristic to take advantage of GB.
- Improve scalability of color choice with the number of decisions.
- Relax SMT stack discipline into multi stack.
- Implement GB in cvc5 [BBB+22].

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Thank you for your attention!

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